Results Interpretation Report

Predicting photostability (T80) of thiophene-based molecules from computed descriptors - Kaggle *molecular-machine-learning* competition

1. Executive summary

- **Small data, many features.** 42 labelled molecules, 143 numeric features, 9 test molecules. Most of the work is about not overfitting.
- **Best validated model:** Ridge (alpha = 1) on just three features (TDOS4.0, NumHeteroatoms, Mass): repeated 5-fold CV MSLE **0.647** (RMSLE 0.804), a **42.9% improvement** over predicting the mean (MSLE 1.133).
- **All-feature models are much worse.** Ridge with alpha = 1 on all 143 features scores 3.578, three times worse than the mean baseline. The best all-feature model (random forest) reaches only 0.821.
- **Two features carry most of the signal:** TDOS4.0 and NumHeteroatoms. Both are negatively related to T80. They are ranked first or near first by random forest, permutation importance and Lasso (average ranks 3.0 and 4.0 out of 143). Mass adds only about 0.008 MSLE on top of those two.
- **Submission:** 70/30 blend of Ridge and a depth-4 random forest on the three features. Leave-one-out MSLE 0.680 versus 0.685 for Ridge alone, so the blend is a marginal, not a meaningful, gain. Predictions for the 9 test molecules lie between 19.8 and 38.0.
- **Honest expectation:** a typical error of about a factor of 2 in T80 (RMSLE ~0.8). With only 42 training and 9 test molecules, the leaderboard score will be noisy.

2. Data

The training file has 42 rows and 146 columns; the test file has 9 rows and the same 146 columns. Every column except Batch_ID, Smiles, T80 (and the derived log target) is numeric, giving **143 features**: molecular descriptors (Mass, LogP, TPSA, ...), electronic properties (HOMO/LUMO, excitation, dipole), and three families of spectra-like columns (T1-T20, S1-S20, O1-O20) plus density-of-states curves (TDOS1.5-4.7, SDOS2.5-5.4).

Missing values: none in the training set. The test file reports 9 missing values in total, none in the three features used for the final model; this equals the number of test rows and is consistent with the test T80 column being empty (inferred from the counts, not inspected directly). That column is never used.

Target distribution: strongly right-skewed. Mean 22.7, median 10.5, standard deviation 26.9, range 1.5 to 103.9, quartiles 5.1 / 10.5 / 30.2. Because the competition metric is MSLE, the models are trained on log(1 + T80), which makes the target much more symmetric and turns plain mean squared error into exactly the competition metric.

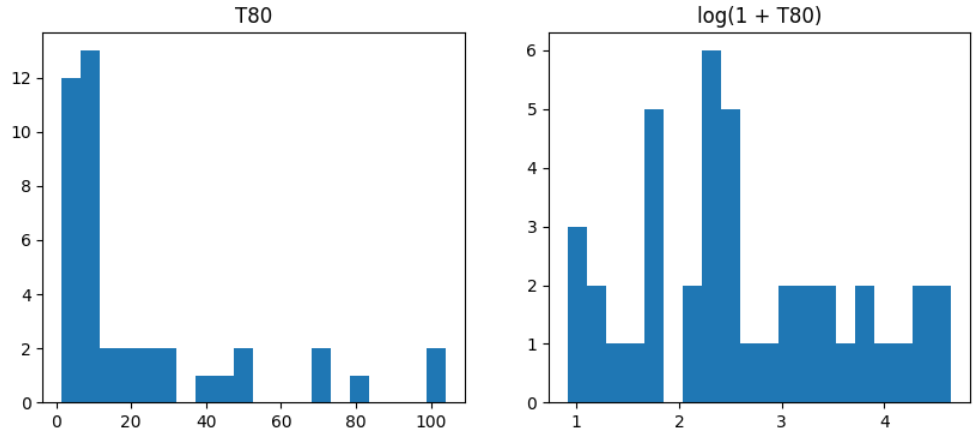


Figure 1. T80 (left) is skewed with a long tail up to ~104; log(1 + T80) (right) is far more balanced.

3. Chemistry check

All 42 SMILES strings parse in RDKit. The provided Mass and NumHeteroatoms columns match RDKit's molecular weight and an independent heteroatom count for every molecule, so the descriptor columns can be trusted.

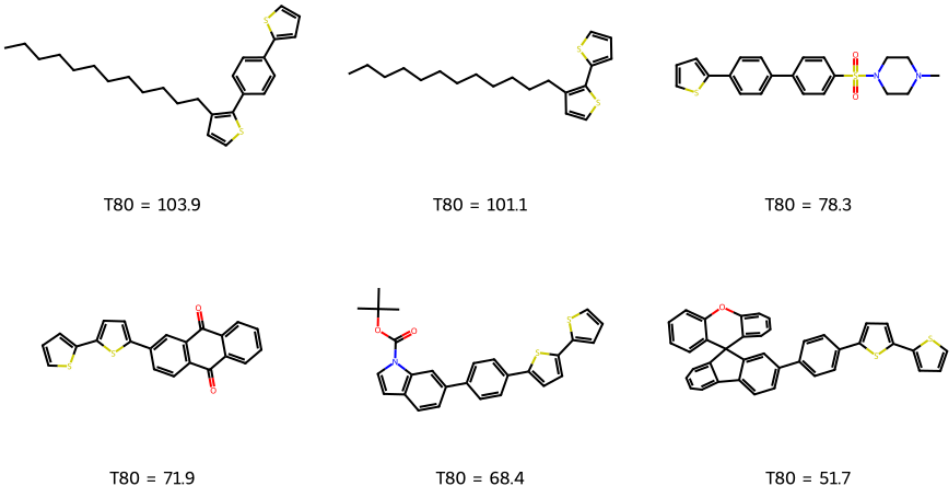


Figure 2. The six most photostable training molecules shown with their T80 (103.9, 101.1, 78.3, 71.9, 68.4, 51.7).

Qualitative reading. The two most stable molecules (T80 about 101-104) are thiophene-phenyl compounds carrying a long alkyl chain and containing only two heteroatoms (the sulfur atoms). The next ones contain sulfonamide, carbonyl or carbamate groups and have four to five heteroatoms. Note that the heteroatom count includes the thiophene sulfur atoms, so it is a crude proxy for how much polar or functional-group content a molecule has; the data show the association but do not explain the mechanism.

4. Target-feature relationships

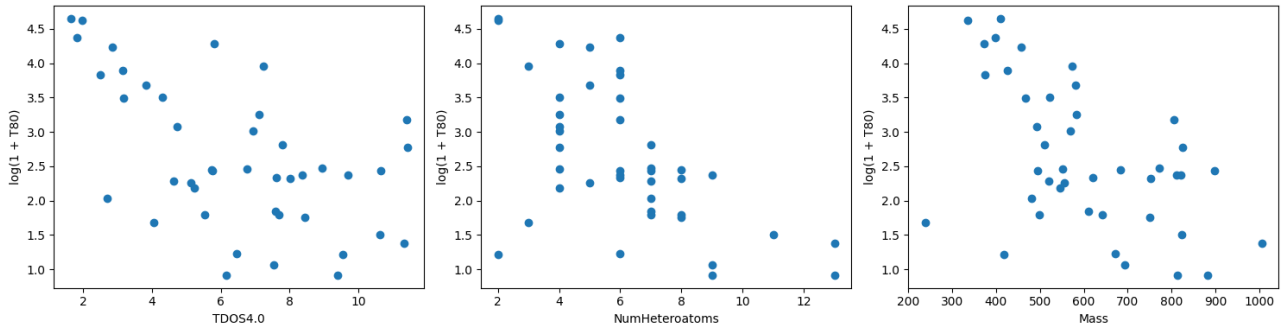


Figure 3. log(1 + T80) against TDOS4.0, NumHeteroatoms and Mass. All three show a negative trend with substantial scatter.

Highest absolute correlations with log(1 + T80) among all features: NumHeteroatoms 0.585, HAcceptors 0.575, TDOS3.9 0.574, TDOS4.0 0.556, then a group of S-columns (S12-S17) at 0.53-0.54. Nothing exceeds 0.59, so no single feature explains the target; the best one explains roughly a third of the variance.

Hypotheses carried forward: (1) more heteroatoms go with lower photostability; (2) a higher density of states near 3.9-4.0 (TDOS) goes with lower photostability; (3) heavier molecules look less stable on their own, but see Section 6 for why Mass behaves differently inside a multi-feature model.

5. Model comparison (cross-validated)

All model selection uses repeated 5-fold cross-validation (5 folds x 3 repeats = 15 scores per model) with MSLE as the metric. Training error is not used for comparison: with 143 features and 42 molecules both models reach a train RMSLE of about 0.34, which says nothing about unseen molecules.

Model	CV MSLE	Std	RMSLE	vs. baseline
Ridge alpha=1, 3 features	0.647	0.225	0.804	+42.9%
Random forest depth 4, 3 features	0.746	0.309	0.864	+34.2%
Random forest, all features	0.821	0.313	0.906	+27.5%
Random forest depth 4, all features	0.829	0.312	0.910	+26.8%
Ridge alpha=100, all features	0.863	0.392	0.929	+23.8%
Mean baseline	1.133	0.315	1.064	0.0%
Ridge alpha=10, all features	1.767	1.619	1.329	-56.0%
Ridge alpha=1, all features	3.578	4.313	1.892	-215.9%

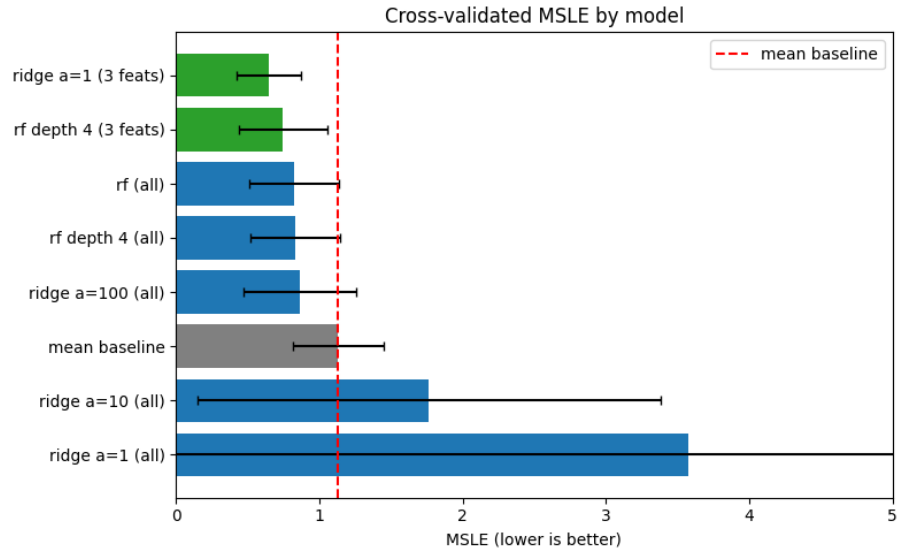


Figure 4. Mean CV MSLE with one-standard-deviation error bars. Green = 3-feature models; dashed line = mean baseline.

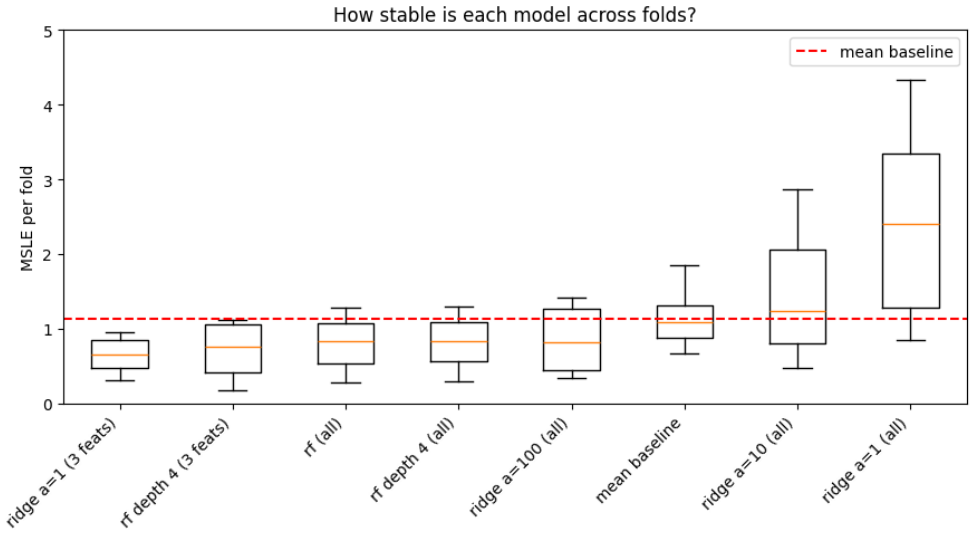


Figure 5. Distribution of the 15 fold scores per model. Weakly regularised all-feature ridge is erratic.

Interpretation.

- **Regularisation and feature count dominate.** Ridge improves steadily as alpha rises on all features (3.58, 1.77, 0.86 for alpha 1, 10, 100) because with more features than samples the unpenalised fit chases noise. Cutting the input to three features does the same job better.

- **Ridge beats the forest on three features** (0.647 vs 0.746), consistent with a roughly linear, monotone relationship in the scatter plots.
- **Differences are within noise.** The fold standard deviation (0.22-0.31) is larger than the gap between the top two models. Ranking the top models is suggestive, not conclusive. The clear findings are the large ones: 3-feature models beat the baseline, and unregularised all-feature ridge loses to it.

6. Which features matter?

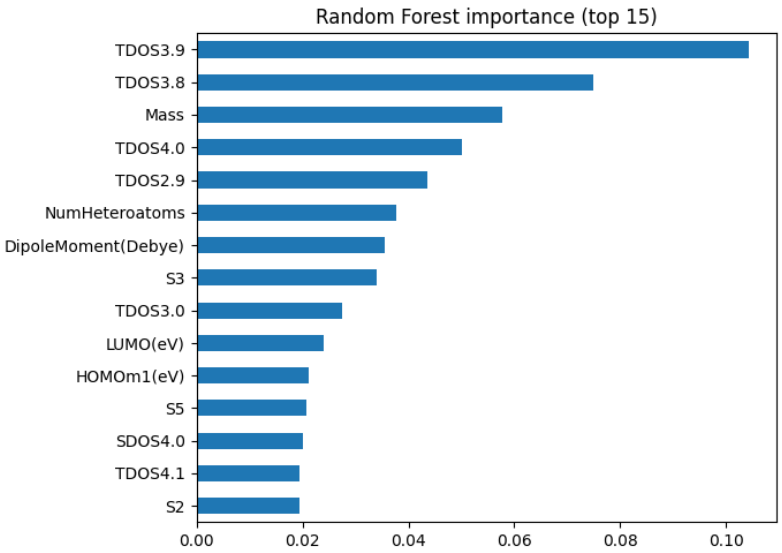


Figure 6. Random forest importance, top 15 features (depth-4 forest, 500 trees).

Three independent views of importance were combined by average rank (lower is better):

Feature	RF rank	Permutation rank	Lasso freq. rank	Average
TDOS4.0	4	1	4	3.0
NumHeteroatoms	6	5	1	4.0
TDOS3.9	1	2	9	4.0
TDOS2.9	5	4	20.5	9.8
Mass	3	3	29.5	11.8
HAcceptors	16	8	14.5	12.8
TDOS3.3	20	14	8	14.0

- **TDOS4.0 is the most consistent feature** (permutation importance 0.034, first of 143). TDOS3.9 and TDOS3.8 rank just as high in the forest, because neighbouring energy bins are strongly correlated and share the importance. The signal is a *region* of the density of states around 3.8-4.0, not one magic bin.
- **NumHeteroatoms is the most stable under Lasso:** it survived 64% of 100 bootstrap refits and has the largest Lasso coefficient (-0.308). Lasso (alpha 0.131) kept 10 of 143 features; apart from the two leaders (TDOS4.0 -0.292, NumHeteroatoms -0.308) the other coefficients are small (|coef| 0.02-0.16).
- **Mass splits the methods.** Tree-based methods rate it third, but Lasso rarely keeps it (rank 29.5), and in Ridge its coefficient is *positive* (+0.277, per standardised feature) even though Figure 3 shows a negative raw trend. Mass and heteroatom count are correlated, so once heteroatoms are in the model, extra mass partly acts as a correction. **Do not read this as 'heavier molecules are more stable'.**

7. Small feature sets

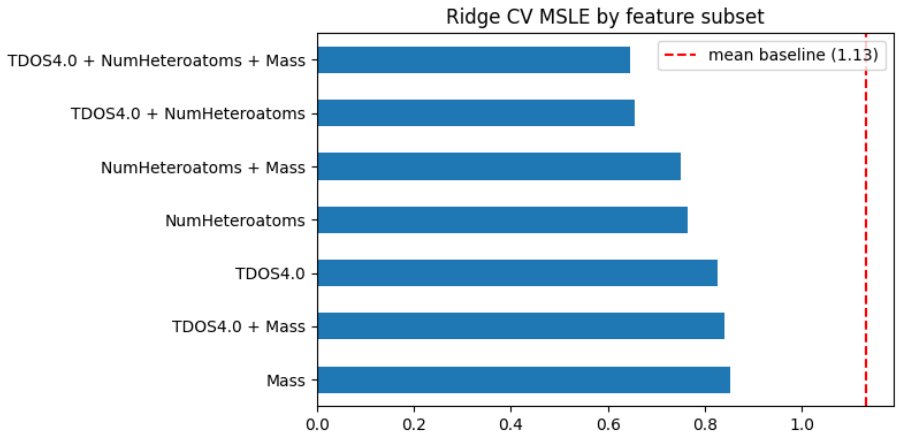


Figure 7. Ridge CV MSLE by subset of the three known features. Dashed line = mean baseline (1.13).

Feature subset (Ridge alpha=1)	CV MSLE
TDOS4.0 + NumHeteroatoms + Mass	0.647
TDOS4.0 + NumHeteroatoms	0.655
NumHeteroatoms + Mass	0.751
NumHeteroatoms	0.764
TDOS4.0	0.827
TDOS4.0 + Mass	0.841
Mass	0.852

- **TDOS4.0 + NumHeteroatoms carries nearly everything** (0.655). Adding Mass improves MSLE by only 0.008, well within noise, so Mass is optional.
- **The two leaders are complementary:** each alone scores 0.76-0.83, together 0.655. They describe different things (electronic structure vs composition).
- **Alpha barely matters** between 0.01 and 3 (0.647-0.652) and degrades from alpha = 10 (0.672) to alpha = 100 (0.827). Alpha = 1 was chosen.
- **Automatic selection inside CV does not match the known features.** SelectKBest + Ridge (alpha 10) scores 1.065 (k=1), 0.988 (k=5) and 0.928 (k=10 or 20). Picking features from 143 with 42 samples is itself noisy; the three known features are a strong prior that data-driven selection cannot reproduce here.
- **Brute-force subsets reach 0.603** (TDOS4.0, NumHeteroatoms, TDOS3.2), better than 0.647 by 0.044. But the shortlist was ranked on all 42 molecules and about 575 subsets were then scored on the same data, so this number is optimistically biased. It is a hypothesis for further testing, not an adopted model.

8. Submission

Range check: all 9 test molecules lie inside the training range for the three features (test ranges: TDOS4.0 2.28-5.12, NumHeteroatoms 2-5, Mass 238-501, versus training 1.63-11.42, 2-13, 238-1005), so the models interpolate rather than extrapolate.

Blend validation (leave-one-out MSLE): mean baseline 1.125, Ridge 0.685, random forest 0.828, 70/30 blend **0.680**.

Final Ridge coefficients (standardised features, log scale): TDOS4.0 -0.566, NumHeteroatoms -0.569, Mass +0.277.

Batch_ID	Ridge T80	RF T80	Blend (submitted)
Test-01	30.59	51.87	35.87
Test-02	30.60	31.63	30.91
Test-03	17.84	25.32	19.82
Test-04	28.41	22.34	26.43
Test-05	41.74	30.39	37.96
Test-06	37.64	33.66	36.40
Test-07	24.15	55.66	31.09
Test-08	24.62	12.55	20.16
Test-09	25.70	17.02	22.73

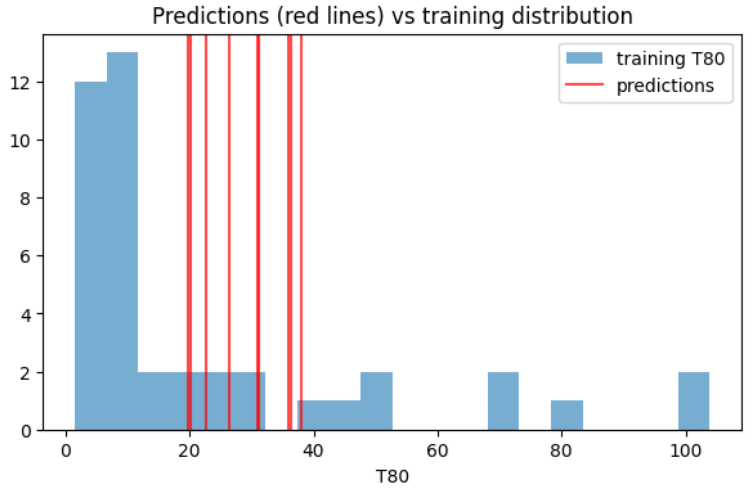


Figure 8. Submitted predictions (red lines) against the training T80 distribution.

- **Predictions are compressed (19.8-38.0)**, as expected for a model with RMSLE ~0.8: it shrinks toward the centre rather than predicting the extremes (1.5 or 100) seen in training.
- **They sit above the training mean (22.7) because the test molecules are small and simple** (at most 5 heteroatoms, mass up to 501). Figure 3 shows that this region of feature space has the highest training T80 values.
- **Model disagreement flags uncertainty.** For Test-01 (Ridge 30.6, forest 51.9) and Test-07 (Ridge 24.2, forest 55.7) the two models differ by a factor of about 1.7-2.3. Those are the predictions to trust least.

9. Symbolic regression (exploratory)

Genetic-programming regression on the same three features gave a CV MSLE of **1.252** (+/- 0.369), worse than the mean baseline (1.133) and about twice the Ridge error (0.647). The formula it found simplifies to 2.654 - X1, where X1 is the standardised NumHeteroatoms: a single straight line. It therefore rediscovered the 'more heteroatoms, lower T80' rule but missed the TDOS4.0 contribution and generalised poorly with 42 points. It was not used for the submission.

10. Limitations and caveats

- **Tiny sample.** 42 training molecules means wide uncertainty on every CV figure (std 0.2-0.4 on MSLE of 0.6-0.9). 9 test molecules means the final score will have high variance.
- **Mild optimism in reported CV.** Alpha, the blend weight and the choice of the three features were tuned or confirmed on the same 42 molecules. The three features come from the competition's hint, which limits but does not eliminate this.
- **Association, not causation.** The analysis shows which descriptors predict T80, not why. No chemical mechanism is established here.
- **Correlated features.** TDOS bins, S/T/O columns and Mass/heteroatoms are strongly related, so individual importance values are unstable; interpret groups, not single columns.
- **No leaderboard score is in the notebook**, so actual competition performance is unknown.

Possible next steps: nested cross-validation for any feature search; testing the 6F subsets with a proper hold-out; adding RDKit fingerprints or group counts (sulfonamide, carbonyl, alkyl chain length) to capture the chemistry visible in Figure 2; reporting prediction intervals.